
MOBILE APP USER ENGAGEMENT & CHURN ANALYSIS

Professional Business Analytics Report

Dataset	Mobile App User Engagement
Records	20,000 users
Variables	15 source fields + derived Engagement Score
Login period	01 Jan 2023 to 01 Jun 2025
Prepared for	Data Analyst Portfolio / Business Review

Purpose: translate user-level behavioral data into clear findings on engagement, subscription mix, churn risk, product usage and potential retention opportunities.

Prepared using the supplied dataset and analysis outputs.

1. Executive Summary

This report evaluates 20,000 mobile-app users across engagement, subscription status, churn-risk scoring, app version, device type, geography, age group, ratings, push-notification interaction and login activity. The source file contains 15 fields and no missing values. The supplied analysis derives an Engagement Score from sessions per day, average session duration, screens viewed and in-app purchases. ■filecite■turn2file1■

Overall engagement is fairly consistent across demographic and geographic segments. Average sessions per day are 7.47, average session duration is 30.45 minutes, and the mean churn-risk score is 0.497. The subscription base is also highly balanced: Premium 33.8%, Free 33.3%, and Trial 32.9%.

The strongest practical opportunity is not simply to compare subscription groups, because their average churn-risk scores are almost identical. Instead, retention work should focus on behavioral segmentation and identify combinations such as high churn risk with low engagement. The current dataset contains a risk score rather than an observed churn outcome, so the analysis should be interpreted as risk profiling, not as a measured churn-rate study.

Monthly login activity is broadly stable across most observed months, with a peak of 742 users in 2023-10. The final month, June 2025, contains only 20 records and should be treated cautiously because it may represent a partial period rather than a genuine collapse in usage.

Headline KPIs

Metric	Value	Interpretation
Users	20,000	Large user-level analytical sample
Avg sessions/day	7.47	Consistent usage frequency
Avg session duration	30.45 min	Substantial session depth
Avg engagement score	245.54	Composite behavioral measure
Avg churn-risk score	0.497	Near midpoint of 0–1 scale
Avg user rating	2.99/5	Moderate satisfaction level

2. Data Overview & Methodology

The analysis is based on the supplied CSV file, which contains 20,000 rows and 15 source columns. The source analysis loads the data with Pandas, validates the structure with `df.info()`, and then creates derived fields and grouped summaries for visualization. ■filecite■turn2file1■

Category	Fields / details
User identity	User ID, Gender, Age
Geography & platform	Country, Device Type, App Version
Engagement	Sessions Per Day, Avg Session Duration Min, Screens Viewed
Messaging	Push Notifications Clicked
Monetization	In App Purchases, Subscription Status
Risk & satisfaction	Churn Risk Score, User Rating
Time	Last Login Date
Derived metric	Engagement Score

Engagement Score construction

The supplied notebook defines Engagement Score as: **(Sessions Per Day × Avg Session Duration Min) + (Screens Viewed × 0.5) + (In App Purchases × 2)**. ■filecite■turn2file1■ This is a custom composite index rather than a standard industry metric, so it is most useful for relative comparison within this dataset.

Age segmentation

Users are grouped into <18, 18–24, 25–34, 35–44, 45–54, 55–64 and 65+ categories using the age bins implemented in the supplied notebook. ■filecite■turn2file8■

Data quality

All 15 source columns are populated for all 20,000 rows, resulting in 0 missing source values. The source fields include 6 object columns, 5 integer columns and 4 floating-point columns. The analysis converts Last Login Date to datetime before creating monthly login summaries. ■filecite■turn2file1■ ■filecite■turn2file2■

3. Descriptive User Profile

The dataset covers users aged 13–64, with a mean age of 38.6. Gender is broadly balanced across the three recorded categories: Other 6,781, Female 6,656, and Male 6,563.

Subscription mix

Subscription	Users	Share	Avg churn risk	Avg engagement	Avg sessions/day
Free	6,653	33.3%	0.499	244.32	7.42
Trial	6,580	32.9%	0.493	244.28	7.53
Premium	6,767	33.8%	0.499	247.96	7.47

The subscription base is unusually balanced. Premium represents the largest group at 33.8%, followed by Free at 33.3% and Trial at 32.9%. This balance creates a useful foundation for comparing user behavior without one category overwhelmingly dominating the sample. The supplied notebook calculates these shares directly from Subscription Status. ■filecite■turn2file8■

Platform mix

Android accounts for 10,009 users (50.0%), while iOS accounts for 9,991 users (50.0%). The platform split is effectively even.

App-version mix

App version	Users	Share
1.0	3,926	19.6%
1.1	3,980	19.9%
1.2	4,084	20.4%
2.0	3,935	19.7%
2.1	4,075	20.4%

4. User Engagement Analysis

Across all users, average sessions per day are 7.47, average session duration is 30.45 minutes, average screens viewed are 26.99, push notifications clicked average 4.46, and in-app purchases average 2.00.

The supplied dashboard visualizes the distributions of session duration, screens viewed, in-app purchases, Engagement Score, Churn Risk Score and User Rating. ■filecite■turn2file0■



Figure 1. Key engagement and risk metric distributions from the supplied analysis.

Engagement Score

The derived Engagement Score has a mean of 245.54, median of 191.94, and standard deviation of 191.26. Its distribution is right-skewed, with the upper quartile at 359.88 and a maximum of 868.94. This suggests that a relatively smaller group of users contributes disproportionately high engagement scores.

Analytical implication

Because the composite score is driven primarily by sessions per day and session duration, it should be treated as an engagement index rather than a direct measure of product value. The score is useful for ranking or segmenting users, but its weighting scheme should be documented and, ideally, validated against a downstream business outcome such as retention, conversion or revenue.

5. Segment-Level Findings

Age group engagement

Age group	Avg sessions/day
<18	7.50
18-24	7.51
25-34	7.40
35-44	7.60
45-54	7.37
55-64	7.48

The highest observed average is the 35–44 group at 7.60 sessions/day, while the 45–54 group is lowest at 7.37. The overall spread is small, indicating that age is not a strong differentiator of session frequency in this sample. The notebook uses the same age-group construction and aggregation. ■filecite■turn2file8■

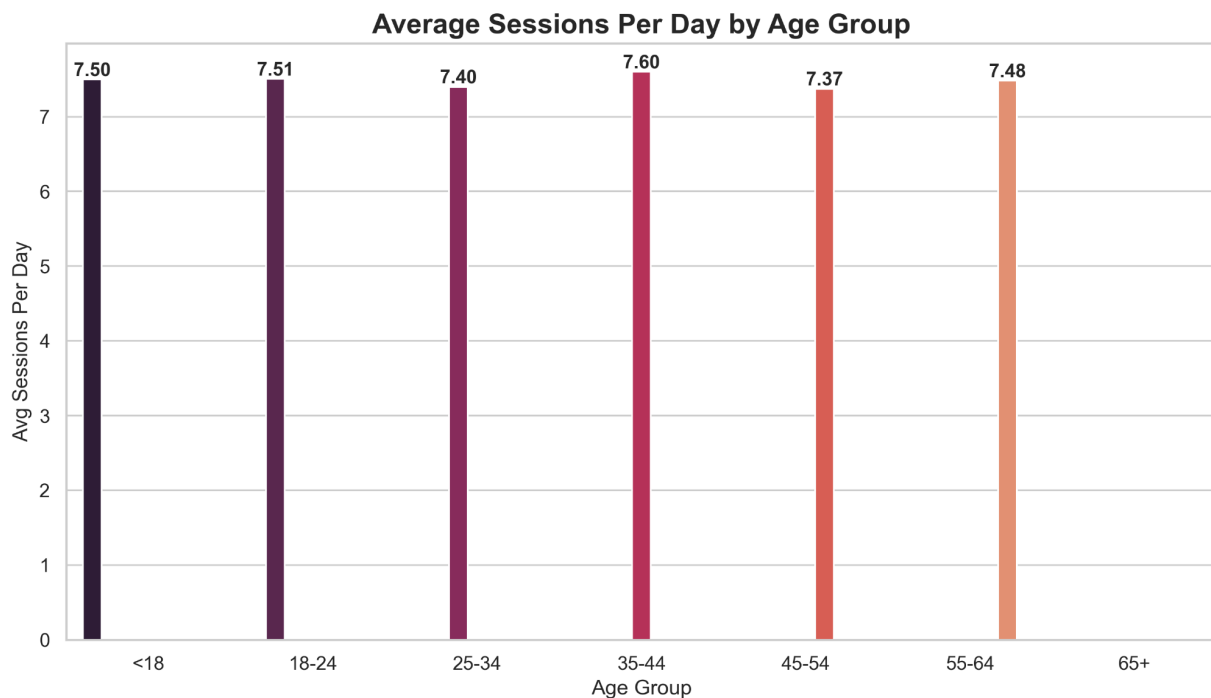


Figure 2. Average sessions per day by age group.

Country engagement

Country	Avg sessions/day
USA	7.53
Brazil	7.51
Germany	7.48
Australia	7.46
Canada	7.45
India	7.45

Country	Avg sessions/day
UK	7.43

The USA has the highest average session frequency at 7.53, while the UK is lowest at 7.43. The difference is only 0.10 sessions per day, so geographic differences appear modest. The supplied notebook explicitly groups Country by Sessions Per Day and sorts the result descending. ■filecite■turn2file7■

6. Churn Risk Analysis

The average Churn Risk Score is 0.497, with a median of 0.50. The score ranges from 0 to 1 and is analyzed as a risk indicator. The supplied notebook uses a violin plot with quartiles to compare score distributions across subscription types. ■filecite■turn2file4■

Subscription	Mean risk	Median	Std. dev.
Free	0.499	0.50	0.289
Trial	0.493	0.50	0.288
Premium	0.499	0.50	0.289

The central finding is that average risk is nearly identical across the three subscription groups. Premium is 0.499, Free is 0.499, and Trial is 0.493. Therefore, subscription status alone does not provide meaningful separation of churn risk in this dataset.

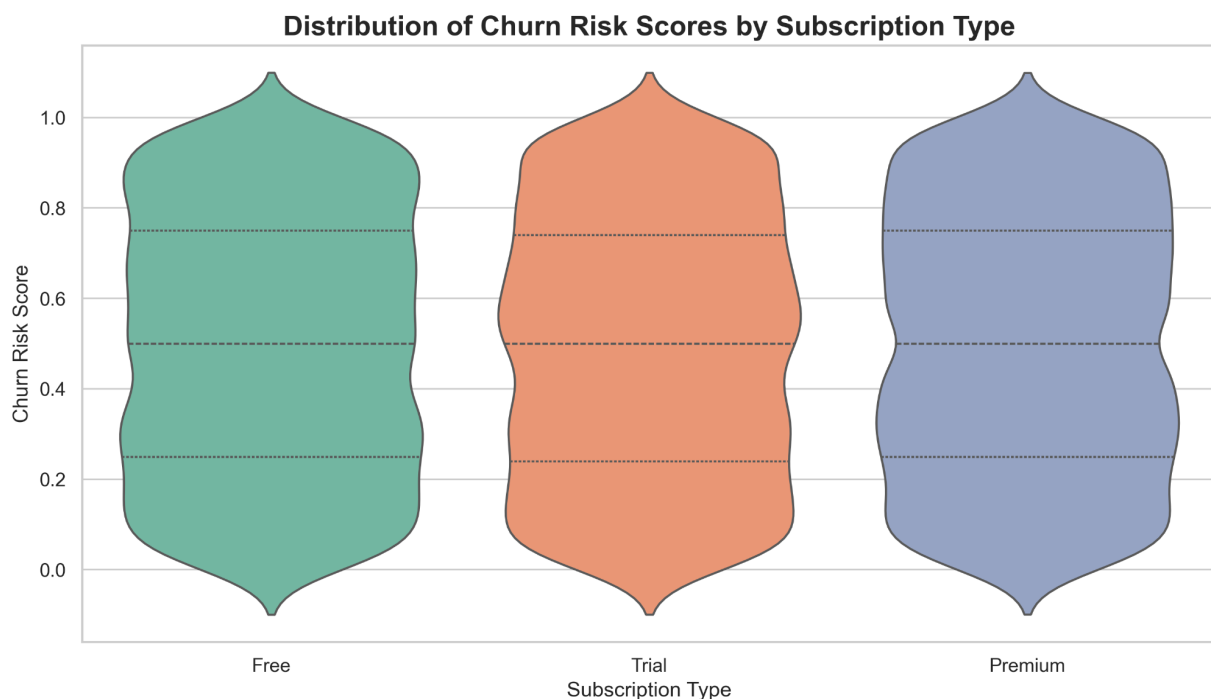


Figure 3. Churn risk score distribution by subscription type.

Behavioral segmentation opportunity

A stronger next step would be to segment users jointly on risk and behavior, for example: **High Risk + Low Engagement**, **High Risk + High Engagement**, **Low Risk + Low Engagement** and **Low Risk + High Engagement**. This would allow retention interventions to be more targeted than a subscription-only strategy.

Important interpretation note

The dataset does not contain an observed binary churn outcome. Consequently, this project cannot establish a measured churn rate, churn drivers or causal relationships. Churn Risk Score should be treated as an existing risk indicator supplied by the dataset.

7. Product Usage & Time Trends

Monthly login activity

Login activity is available from 01 Jan 2023 through 01 Jun 2025. Monthly counts were created by converting Last Login Date to datetime, extracting YYYY-MM periods, counting records and sorting chronologically, matching the supplied notebook methodology. ■filecite■turn2file2■

The highest monthly count is 742 in 2023-10. Most full months fall within a relatively narrow range around 625–742 users. June 2025 has only 20 records, which should be flagged as a likely partial-period observation until the business confirms data completeness.

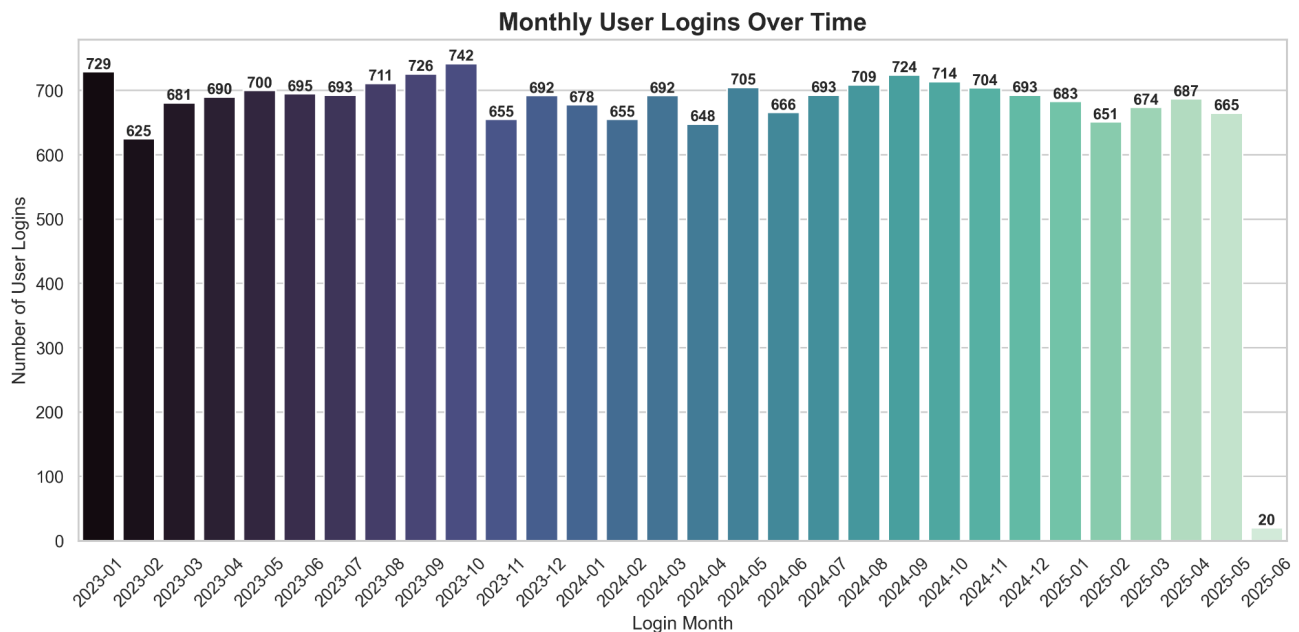


Figure 4. Monthly user login activity. The final month should be interpreted cautiously.

Push notification interactions

Users average 4.46 push-notification clicks, with a median of 4. The supplied analysis visualizes the distribution using a histogram with a KDE overlay. ■filecite■turn2file6■

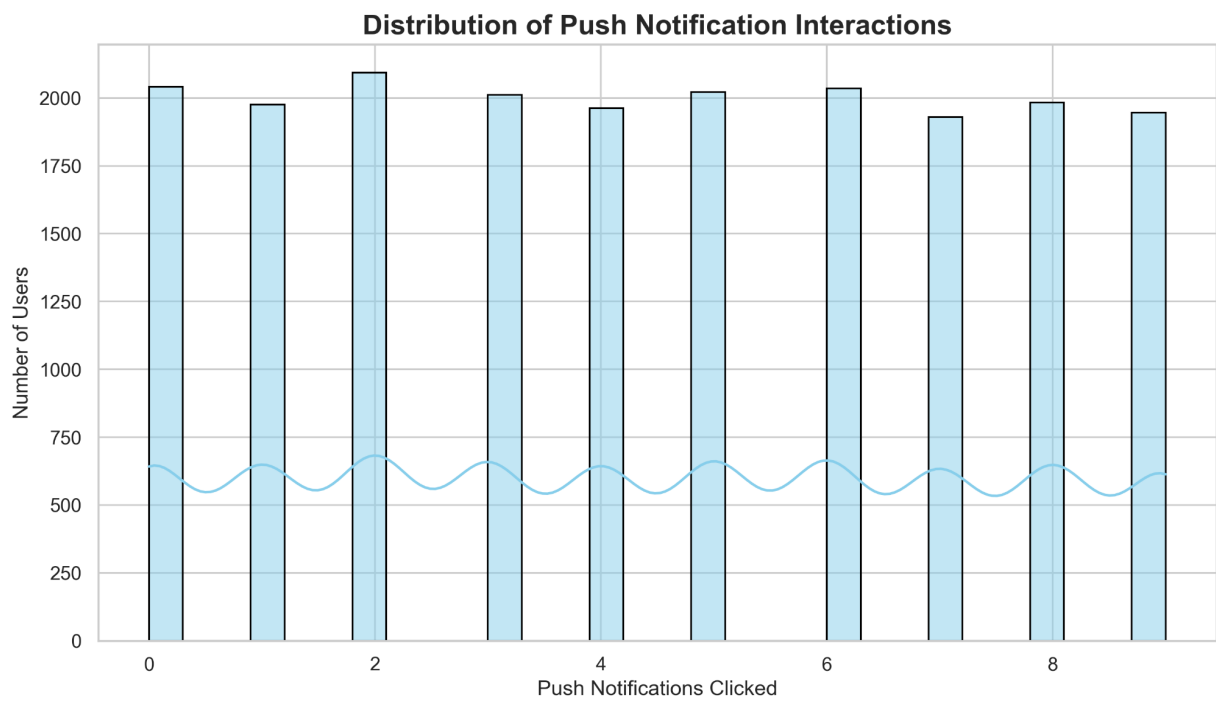


Figure 5. Distribution of push notification interactions.

8. Device & App-Version Analysis

Device type

Average session duration is 30.57 minutes on Android and 30.33 minutes on iOS, a difference of only 0.24 minutes. The supplied notebook calculates this comparison by grouping Avg Session Duration Min by Device Type. ■filecite■turn2file9■

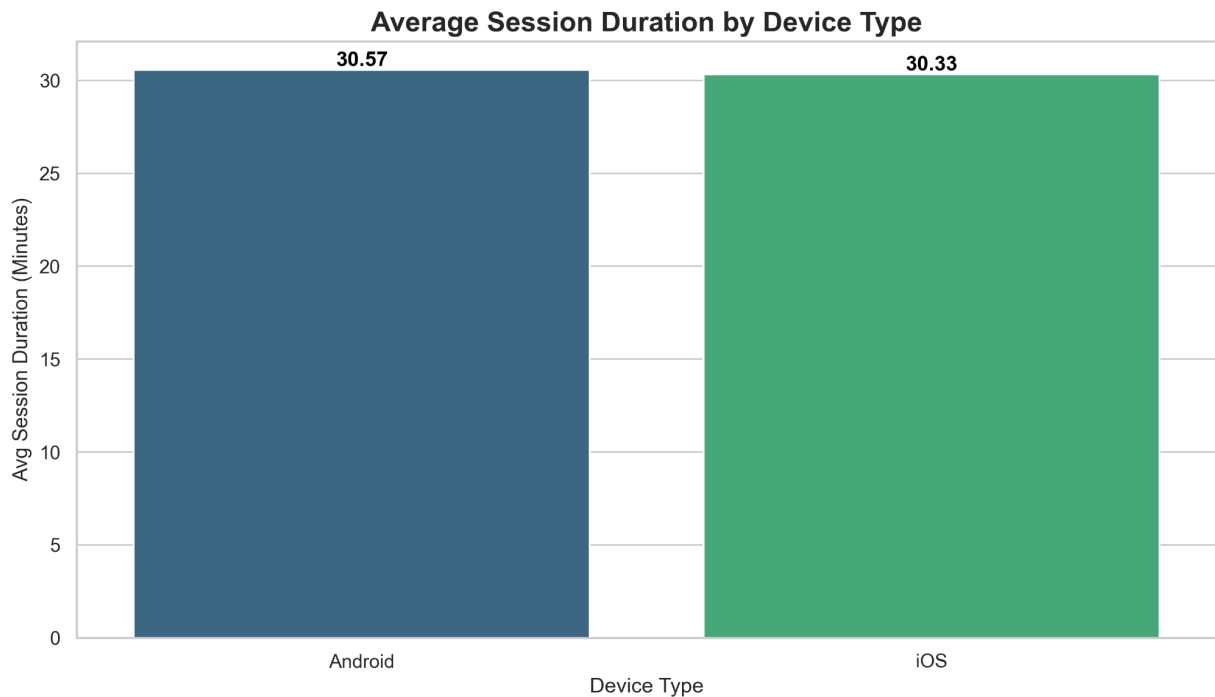


Figure 6. Average session duration by device type.

App version distribution

Version 1.2 has the largest user count at 4,084, followed by version 2.1 at 4,075. The smallest displayed base is version 1.0 at 3,926. The supplied notebook counts users by App Version and sorts versions numerically.

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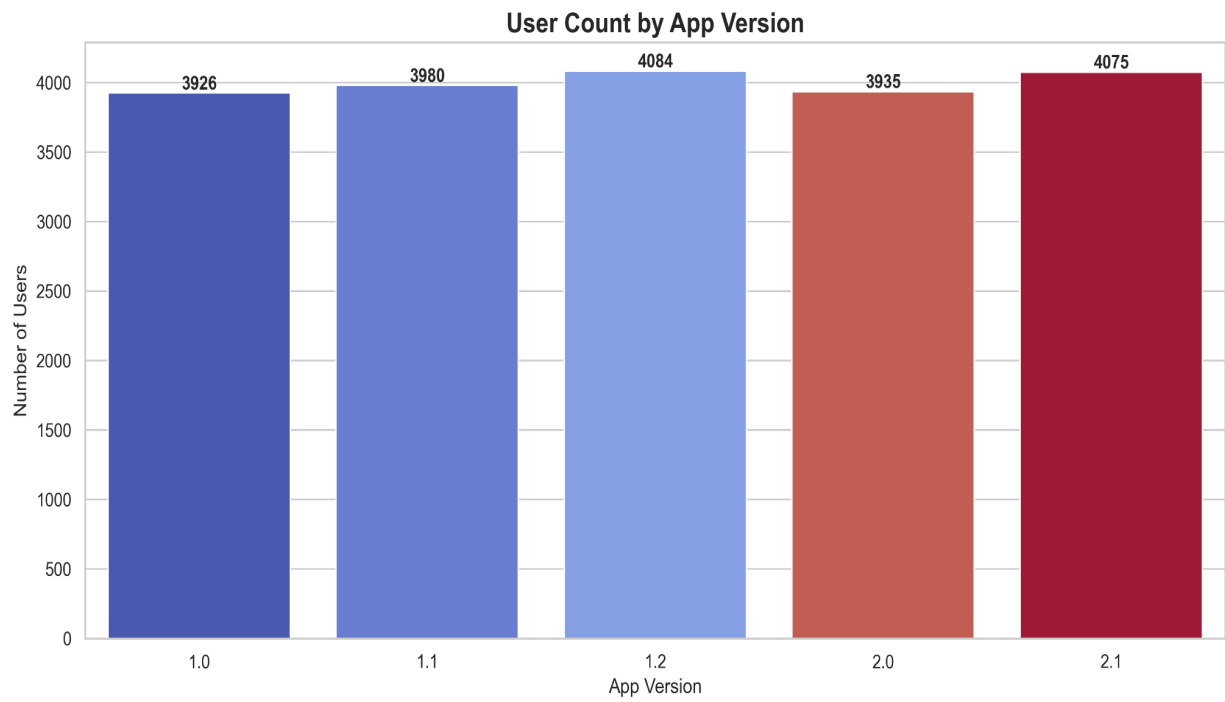


Figure 7. User count by app version.

9. Satisfaction & Relationship Analysis

User ratings

Average user rating is 2.99/5, with a median of 3.0. The supplied dashboard includes the rating distribution as one of the six core user metrics. ■filecite■turn2file0■

Correlation screening

Variable	Correlation with churn risk
Engagement Score	-0.004
In App Purchases	-0.004
Sessions Per Day	-0.002
Avg Session Duration Min	-0.001
User Rating	0.001
Push Notifications Clicked	0.004
Screens Viewed	0.008

The correlation results are notable because every listed behavioral variable has a correlation with Churn Risk Score close to zero. The largest absolute relationship among the tested variables is Screens Viewed at 0.008, which is still negligible in linear terms. This suggests that the supplied risk score is not linearly explained by any single measured behavior in this dataset.

Engagement Score is strongly associated with its two dominant components, average session duration ($r=0.672$) and sessions per day ($r=0.650$). This is expected because both variables are explicitly included in the score formula. It should therefore not be interpreted as an independent validation of the engagement construct.

Analytical caution

Correlation does not establish causation. In addition, the Engagement Score is mathematically constructed from several of the same variables being compared against it. Future analysis should use independent outcomes, time-based features and, where appropriate, statistical tests or predictive models.

10. Business Recommendations

1. Build a behavioral retention segment

Prioritize users who combine high Churn Risk Score with low Engagement Score. These users represent the clearest candidate group for targeted retention campaigns, product education, personalized messaging or incentive testing.

2. Optimize Trial-to-Premium conversion

Trial users represent 32.9% of the base, or 6,580 users. Analyze their behavior near the end of the trial period and compare converters versus non-converters using sessions, session duration, screens viewed, ratings and notification engagement.

3. Validate the churn-risk score

Because the current risk score has almost no linear correlation with the observed engagement variables, the business should validate whether the score predicts a real downstream event such as inactivity, cancellation or subscription downgrade. If an actual churn label becomes available, evaluate AUC, precision, recall, calibration and lift by risk band.

4. Investigate the final login period

June 2025 contains only 20 login records versus hundreds in preceding months. Confirm whether this is a partial data extract, reporting cutoff or genuine product event before using it in trend decisions.

5. Monitor app-version performance

Compare engagement, ratings and churn-risk distributions by app version rather than tracking adoption alone. This can identify releases associated with better or worse user experiences.

6. Measure notification effectiveness

Use push-notification clicks as an exposure/response feature and test whether targeted notification strategies lead to higher subsequent engagement or retention. The current descriptive distribution alone cannot establish effectiveness.

7. Move from descriptive to decision analytics

The next version of the project should add SQL-based business questions and a Power BI dashboard. This would turn the analysis into a stronger end-to-end Data Analyst portfolio project covering data extraction, transformation, analysis, visualization and business communication.

11. Limitations & Next-Step Analysis

Current limitations

- No observed churn/cancellation label is included, so actual churn cannot be measured.
- The Engagement Score uses custom weights and has not been validated against a business outcome.
- Monthly login counts are record counts based on Last Login Date, not necessarily unique active users per month.
- June 2025 appears unusually low and may be a partial period.
- Descriptive comparisons do not establish causality.
- The analysis does not currently include cohort retention, trial conversion, revenue or lifetime value.

Recommended next phase

1. Create a SQL layer for repeatable business queries.
2. Add actual churn or inactivity labels if available.
3. Create retention cohorts and monthly active-user metrics.
4. Compare Trial → Premium conversion by engagement segment.
5. Build a Power BI dashboard with slicers for country, device, subscription, age group and app version.
6. Test churn-risk performance against observed outcomes.
7. Translate the strongest findings into quantified business actions.

Conclusion

This analysis provides a solid descriptive view of mobile-app user behavior. The most important takeaway is that engagement and churn risk should be approached through behavioral segmentation rather than simple demographic or subscription comparisons. The balanced subscription mix, stable session activity and nearly identical device-level session durations suggest a broadly consistent user experience, while the churn-risk score presents an opportunity for deeper validation and targeted retention analysis.

Appendix A. Core Metrics

Metric	Mean	Median	Min	Max
Sessions Per Day	7.47	7.00	1.00	14.00
Avg Session Duration Min	30.45	30.52	1.00	60.00
Screens Viewed	26.99	27.00	5.00	49.00
Push Notifications Clicked	4.46	4.00	0.00	9.00
In App Purchases	2.00	2.00	0.00	4.00
Engagement Score	245.54	191.94	4.13	868.94
Churn Risk Score	0.50	0.50	0.00	1.00
User Rating	2.99	3.00	1.00	5.00

Appendix B. Source Analysis Outputs

The report incorporates the supplied analysis outputs covering key engagement metrics, monthly logins, subscription distribution, churn-risk distribution, age-group activity, country engagement, app-version usage, device-level session duration and push-notification interactions. The supplied analysis documents the corresponding Pandas groupings, derived metrics and visualizations. [filecite turn2file0](#) [filecite turn2file2](#) [filecite turn2file7](#) [filecite turn2file8](#) [filecite turn2file9](#)

Source: Mobile App User Engagement.csv and the accompanying Mobile App User Engagement analysis notebook/PDF.